Proof-Route Alignment for the Periodic Overlap Dichotomy (de las Cuevas–Cirac–Schuch–Pérez-García)

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1 Scope

This note records, in mathematical terms, where the formal development of the periodic overlap dichotomy chooses a proof route that differs from the appendix argument of [1, Appendix A], and where its statements restrict the source.[1]

The source proposition is *equal-or-orthogonal-generalized*.[2] None of the deviations below is unfaithful in the smuggled-hypothesis sense: each formal statement is a faithful (or explicitly restricted) version of the source, and the route substitutions are mathematically equivalent.

2 Different-period decay via the peripheral spectrum

Source.[3] For periods $m_a \neq m_b$, the paper blocks at $p = \mathrm{lcm}(m_a, m_b)$ and uses Lemma *bdcf*: the blocks $P_u A^{(p)}$ and $Q_v B^{(p)}$ are non-repeated normal tensors. Applying the one-site translation operator to $Q_v B^{(p)}$ versus $Q_{v+1} B^{(p)}$ yields a contradiction unless $\lim_N |\langle V_N(A)|V_N(B)\rangle| = 0$.

Lean route.[4] The same obstruction is realized spectrally. For a tensor A, write $\mathcal{E}_A$ and $\mathcal{E}_A^*$ for the transfer map and its adjoint, defined by

$$\mathcal{E}_A(Y) = \sum_i A^i Y A^{i\dagger}, \qquad \mathcal{E}_A^*(Y) = \sum_i A^{i\dagger} Y A^i. \tag{1}$$

[1] Formal files: TNLean/MPS/Periodic/Overlap/. Local source: Papers/1708.00029/main.tex. Paper-level umbrella: issue #619; proposition-level subtracking: issue #81 (self-overlap limit and non-matching decay, issue #448; sector-match propagation and gauge assembly, issue #450).

[2] Stated in [1], lines 589–609, and proved in Appendix A, lines 903–1118.

[3] [1], Appendix A, lines 917–950.

[4] Formal declaration: periodicOverlap˙tendsto˙zero˙of˙ne˙period in Case1.lean.

If $D_a = D_b$ and $B^i = \zeta\, X A^i X^{-1}$, the comparison is

$$\mathcal{E}_B(Y) = \zeta\bar{\zeta}\, X \mathcal{E}_A(X^{-1} Y X^{-\dagger}) X^{\dagger}. \tag{Case 1}$$

Eigenvalue uniqueness for an irreducible completely positive map ([3], Theorem 6.3) forces $|\zeta| = 1$, so $\mathcal{E}_A$ and $\mathcal{E}_B$ have equal peripheral spectra. An m-periodic tensor has peripheral spectrum $\{e^{2\pi i r/m} : 0 \le r < m\}$, so equality forces $m_a = m_b$. Thus different periods exclude gauge-phase equivalence, and *equal-or-orthogonal* (the irreducible trace-preserving overlap dichotomy) gives the decay. This substitutes the paper's lcm-blocking + translation argument; it is complete and proved.

3 Sector non-repetition via the blocked fixed-point structure

Source.[5] For a periodic block, $\mathcal{E}_A$ is irreducible with peripheral spectrum $\{\omega^r\}$, $\omega = e^{2\pi i/m}$. The blocked map $\mathcal{E}_A^m$ then has 1 as its only modulus-one eigenvalue (multiplicity m), with fixed-point set exactly $\{P_u \Lambda_A P_u\}_u$, while the adjoint $\mathcal{E}_A^{*m}$ has fixed points $\{P_u\}_u$. If, for $u \neq v$, a gauge-phase equivalence $C_u^{\mathbf{i}} = e^{i\xi} U C_v^{\mathbf{i}} U^{\dagger}$ held with $U = P_u U P_v$ ($UU^{\dagger} = P_u$, $U^{\dagger}U = P_v$), then

$$\begin{aligned}
\mathcal{E}_A^m(U) &= \sum_{\mathbf{i}} C_u^{\mathbf{i}} U C_v^{\mathbf{i}\dagger} && (2)\\
&= e^{i\xi} U \sum_{\mathbf{i}} C_v^{\mathbf{i}} C_v^{\mathbf{i}\dagger} && (3)\\
&= e^{i\xi} U. && (\mathcal{E}_{C_v}(P_v) = P_v)
\end{aligned}$$

so U is a modulus-one eigenvector of $\mathcal{E}_A^m$. As the only such eigenvalue is 1 with the diagonal fixed points $\{P_w \Lambda_A P_w\}$ and $U = P_u U P_v$ is off-diagonal for $u \neq v$, this is a contradiction.

Lean status.[6] The lemma statement is the faithful non-repetition fact: distinct cyclic sectors (orthogonal projectors $P_u P_v = 0$) are not gauge-phase equivalent. An earlier draft proposed to discharge it by a "BDCF converse" (orthogonal sectors are eventually linearly independent, so a gauge-phase equivalence would force two sector matrix-product-vector families to be proportional). That is *not* the source argument. The faithful route is the spectral one above, which reuses the peripheral-spectrum / fixed-point machinery already invoked in the different-period proof.[7] The docstrings have been realigned to the source argument; the remaining work is to port it.

Signature correction (periodicity is load-bearing). An adversarial proof-attempt found that the lemma was *underspecified*: with only the

[5] Lemma *bdcf* in [1], lines 404–423.

[6] Formal declaration: not_gaugePhaseEquiv_of_orthogonal_cyclicSector_traces in SelfOverlap.lean; the proof is still open.

[7] Formal declaration: period_eq_of_gaugePhaseEquiv_of_isPeriodic in Case1.lean.

bare cyclic-projection data (orthogonal P_k summing to 1, the adjoint shift $\mathcal{E}_A^*(P_{k+1}) = P_k$, blocked-letter commutation, the trace formula, and orthogonality), the conclusion is *false* — a reducible $A = B \oplus B$ (for an irreducible period-m block B) admits the same cyclic-projection data yet has gauge-phase-equivalent distinct sectors. The spectral argument genuinely requires A to be a periodic (irreducible) block: that is what makes $\mathcal{E}_A$ irreducible with the stated peripheral spectrum and the sectors primitive. The hypothesis hP : IsPeriodic m A was therefore added to the lemma signature (it is supplied at the unique call site sectorBlocks_not_gaugePhaseEquiv_of_ne). The sorry now sits on a *true* statement; the remaining work is to port the spectral proof.

4 Sector-match case: propagation and the phase assembly

Source.[8] Given one matched pair, applying the translation operator T^l to the blocked equation *eq:Nm* and invoking Theorem 2.10 of [2] produces, at each offset, a phase and a unitary $U_{\tilde{v}+l} = P_{\tilde{u}+l} U_{\tilde{v}+l} Q_{\tilde{v}+l}$, so the offset $v - u = q$ is constant. Then each blocked product F_u becomes injective after N_0 repetitions, with a right inverse Ω_u; contracting the concatenated products with the Ω_u recovers per-site proportionality $A_u^i = \kappa_v \, e^{i\eta/m} \, B_v^i$. The phase bookkeeping is load-bearing: $\prod_v \kappa_v = 1$ (*eq:prodkappaprop*) and $|\kappa_v| = 1$ from $\| \sum_i A_u^{i\dagger} A_u^i \| = 1$, so $\kappa_v = e^{i\theta_v}$ with $\sum_v \theta_v = 0$; choosing ϕ_v with $\theta_v = \phi_v - \phi_{v+1}$ telescopes the per-sector phases into a single global phase $\xi = \eta/m$ and a single global unitary $U = \sum_u e^{i\phi_{u+q}} P_u U_{u+q} Q_{u+q}$, giving $A^i = e^{i\xi} U B^i U^\dagger$.

Lean status.[9] The statements track this two-stage structure: the T^l + Theorem 2.10 staircase, followed by the Ω_u contraction and the $\kappa/\theta/\phi$ phase assembly. The top-level theorem now invokes these assembly lemmas and is proved modulo the two remaining leaf obligations. The docstrings have been realigned to cite the source equation labels and line ranges (replacing earlier published-version labels such as "A.8" and "A.14–A.18" that do not occur in the local source), and the $\kappa/\theta/\phi$ phase assembly is now stated explicitly as load-bearing.

Missing infrastructure (identified by adversarial proof-attempts). The two open Case-3 leaves reduce to a common gap: IsCyclicSectorDecomp exposes only *blocked-letter* data (the commutation $P_k \, (\bar{A})_i = (\bar{A})_i \, P_k$ and the adjoint shift $\mathcal{E}_A^*(P_{k+1}) = P_k$), but the appendix argument needs *single-site* structure that is currently absent from the repository (and from Mathlib):

- *Single-site off-diagonal decomposition* $A^i = \sum_u P_u A^i P_{u+1}$ (*eq:Auprop*, line 1043). This is the bridge from the corner-compressed sectors back

[8] [1], Appendix A, lines 961–1117; the initial blocked equation *eq:Nm* appears at lines 978–984.

[9] Formal declarations: sectorMatch_propagation, sectorTensor_proportional_of_blockedMatch, repeatedBlocks_of_blockedSectorGaugePhase, and periodicOverlap_gaugeEquiv_of_sector_match in Case3.lean.

to the global matrices A^i, required to even *state* the contraction in repeatedBlocks˙of˙blockedSectorGaugePhase. Adding it (as a field of IsCyclicSectorDecomp or a derived lemma) is the key enabling step.

- *Single-site sector-overlap translation invariance*: that $\|\langle V_N(A_{u+1})|V_N(B_{v+1})\rangle\|$ and $\|\langle V_N(A_u)|V_N(B_v)\rangle\|$ share an $N \to \infty$ limit, obtained by applying the dual single-site shifts $\sum_i A_i^\dagger P_{u+1} A_i = P_u$ and $\sum_i B_i^\dagger Q_{v+1} B_i = Q_v$ inside the paired trace sum. This is the one missing lemma for sectorGaugePhaseEquiv˙succ˙of˙cyclicTransport; given it, that leaf closes via the proved gaugePhaseEquiv˙of˙overlap˙norm˙tendsto˙one˙of˙irreducible˙TP and the proved sector primitivity. There is no one-physical-site offset reblocking equivalence on configurations in the repo (only the $m \cdot n$ regrouping).
- *Per-sector corner unitaries*: upgrading the compression LinearEquiv φ_u to a multiplicative $*$-isomorphism with a unitary intertwiner $U_v = P_u U_v Q_v$ on Fin D (*eq:Cvprop*); and the m-factor cyclic Ω_u-contraction with the η bookkeeping (the two-site tensor˙proportional is its $m = 2$ shadow) plus the $\kappa/\theta/\phi$ telescoping into the global unitary $U = \sum_u e^{i\phi_{u+q}} P_u U_{u+q} Q_{u+q}$.

These are the remaining obligations for sectorGaugePhaseEquiv˙succ˙of˙cyclicTransport and repeatedBlocks˙of˙blockedSectorGaugePhase; each is a focused infrastructure development rather than a one-step discharge.

5 Scope restriction: independence without spanning

Source.[10] As a consequence of the proposition and Lemma *Lem1t*, for $N \geq N_0$ the non-zero members of $\{|V_N(A_j)\rangle\}_{j\in J}$ are linearly independent *and* span $|V_N(A)\rangle$; the spanning half is what "justifies the name basis of periodic vectors".

Lean statement.[11] Only the independence half is stated; the spanning clause is dropped. The basis condition is encoded directly as the pairwise non-repetition hypothesis hNonrep together with a common-period divisibility hDiv, rather than derived from a basis of periodic tensors, and the *Lem1t* almost-orthonormal-implies-independent lemma is not used. Restoring the spanning clause and routing through *Lem1t* is tracked under issue #81.

References

[1] G. de las Cuevas, J. I. Cirac, N. Schuch, and D. Pérez-García, *Irreducible forms of matrix product states: Theory and applications*, J. Math. Phys. **58**, 121901 (2017), arXiv:1708.00029.

[10] [1], lines 604–608 and Lemma *Lem1t*, lines 511–519. The phrase about the name "basis of periodic vectors" appears at line 611.

[11] Formal declaration: periodicBasis˙eventuallyLinearlyIndependent in Dichotomy.lean; blueprint label `thm:periodic_basis_eventually_li`.

[2] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, *Matrix product states and projected entangled pair states: Concepts, symmetries, theorems*, Ann. Phys. **378**, 100 (2017) ("Ci15" in arXiv:1708.00029).

[3] M. M. Wolf, *Quantum Channels and Operations: Guided Tour*, unpublished lecture notes (2012).